Grade 11 Practice

Standard N.CN.1

Computation

1. Solve: i^2 = ___ **-1**

2. Solve: i^3 = ___ **-i**

3. Solve: i = ___ **1**

4. Write $\sqrt{-9}$ in terms of i. ___ **3i**

Word problems

5. Write $\sqrt{-16}$ as a complex number.
Answer: 4i

6. What is the real part of 3 + 4i?
Answer: 3

Grade 11 Practice

Standard N.CN.2

Computation

1. Solve: (2 + 3i) + (1 - i) = ___ **3 + 2i**

2. Solve: (1 + i)(1 - i) = ___ **2**

3. Solve: (2 + i)(2 - i) = ___ **5**

4. Solve: (3 + 2i) - (1 + 4i) = ___ **2 - 2i**

Word problems

5. An impedance is (4 + 2i) plus (1 + i). What is the total?
Answer: 5 + 3i

6. Solve: i(3 + i) = ___
Answer: -1 + 3i

Grade 11 Practice

Standard N.CN.7

Computation

1. Solve $x^2 + 1 = 0$. x = ___ **$\pm i$**

2. Solve $x^2 + 4 = 0$. x = ___ **$\pm 2i$**

3. Solve $x^2 + 9 = 0$. x = ___ **$\pm 3i$**

4. Solve $x^2 + 2 = 0$. x = ___ **$\pm i\sqrt{2}$**

Word problems

5. What is the sum of the roots of $x^2 + 1 = 0$?
Answer: 0

6. What is the product of the roots of $x^2 + 4 = 0$?
Answer: 4

Grade 11 Practice

Standard A.SSE.3

Computation

1. Write $x^2 + 6x + 5$ in vertex form. ___ **$(x + 3)^2 - 4$**

2. What is the minimum value of $x^2 - 4x + 7$? ___ **3**

3. What is the sum of the roots of $x^2 - 5x + 6 = 0$? ___ **5**

4. What is the product of the roots of $2x^2 + 8x + 6 = 0$? ___ **3**

Word problems

5. A ball's height is $h = -16t^2 + 32t$, where t is seconds. What is its maximum height?
Answer: 16 feet

6. The function $y = -x^2 + 6x$ has a maximum value. What is it?
Answer: 9

Grade 11 Practice

Standard A.APR.2

Computation

1. Let $p(x) = x^2 - 3x + 2$. What is $p(1)$? ___ **0**

2. Let $p(x) = x^3 - x$. What is $p(2)$? ___ **6**

3. Is $x - 2$ a factor of $x^2 - 4$? Write yes or no. **yes**

4. What is the remainder when $x^2 + 1$ is divided by $x - 1$? ___ **2**

Word problems

5. Is $x + 1$ a factor of $x^3 + 1$? Write yes or no.
Answer: yes

6. What is the remainder when x^3 is divided by $x - 2$?
Answer: 8

Grade 11 Practice

Standard A.REI.2

Computation

1. Solve $2/x = 4$. $x =$ ___ **$1/2$**

2. Solve $6/(x + 1) = 3$. $x =$ ___ **1**

3. Solve $x/3 = 2/x$. Write both solutions. **$\pm\sqrt{6}$**

4. Solve $(x + 2)/x = 3$. $x =$ ___ **1**

Word problems

5. A rate is $12/x = 3$. What is x?
Answer: 4

6. A tank fills in $6/x$ hours, and the time is 2 hours. What is x?
Answer: 3

Grade 11 Practice

Standard A.REI.7

Computation

1. Solve the system $y = x$ and $y = x^2$. Write the solutions. **$(0, 0)$ and $(1, 1)$**

2. Solve the system $y = 2$ and $y = x^2$. Write the x-values. **$\pm\sqrt{2}$**

3. Solve the system $y = x + 2$ and $y = x^2$. Write the solutions. **$(2, 4)$ and $(-1, 1)$**

4. Solve the system $y = 0$ and $y = x^2 - 4$. Write the x-values. **± 2**

Word problems

5. A path $y = x^2$ reaches ground level $y = 0$. Where does it meet the ground?
Answer: $(0, 0)$

6. A circle $x^2 + y^2 = 25$ meets the line $y = 0$. What are the x-values?
Answer: ± 5

Grade 11 Practice

Standard A.REI.11

Computation

1. Find where $y = 2x$ meets $y = x + 3$. Write (x, y). **$(3, 6)$**

2. Find the x-values where $y = x^2$ meets $y = 4$. **± 2**

3. Find where $y = 3x - 1$ meets $y = x + 5$. Write (x, y). **$(3, 8)$**

4. Find where $y = 1/x$ meets $y = x$. Write the solutions. **$(1, 1)$ and $(-1, -1)$**

Word problems

5. Costs $y = 10x$ and $y = 5x + 50$ are equal at what x?
Answer: $x = 10$

6. A parabola $y = x^2 - 4$ meets the line $y = 0$. What are the x-values?
Answer: ± 2

Grade 11 Practice

Standard F.IF.6

Computation

1. Find the average rate of change of $f(x) = 2x + 3$ on $[1, 4]$. ___ **2**

2. Find the average rate of change of $f(x) = x^2$ on $[0, 2]$. ___ **2**

3. Find the average rate of change of $f(x) = x^2$ on $[1, 3]$. ___ **4**

4. Find the average rate of change of $f(x) = 5$ on $[0, 10]$. ___ **0**

Word problems

5. A distance is $d = 3t$, where t is hours. What is the average rate of change from $t = 0$ to $t = 4$?
Answer: 3

6. A temperature is 20 at $t = 0$ and 40 at $t = 4$. What is the average rate of change per hour?
Answer: 5

Grade 11 Practice

Standard F.IF.7

Computation

1. Where is the vertex of $y = |x|$? Write the ordered pair. **(0, 0)**

2. What is the y-intercept of $y = x^2 + 2$? ___ **2**

3. Where is the vertex of $y = -x^2$? Write the ordered pair. **(0, 0)**

4. What is the domain start of $y = \sqrt{x}$? ___ **0**

Word problems

5. Where is the vertex of $y = |x - 3|$? Write the x-value.
Answer: 3

6. What is the y-intercept of $y = 2$?
Answer: 1

Grade 11 Practice

Standard F.IF.8

Computation

1. Write $y = x^2 + 6x + 5$ in vertex form. ___ **$(x + 3)^2 - 4$**

2. Write y = 2^(3x) with base 8. ___ **8**

3. Write y = 1.05^(2t) with base 1.1025. ___ **1.1025**

4. Factor $y = x^2 - 4$. ___ **$(x - 2)(x + 2)$**

Word problems

5. A balance is y = 1000(1.05)^t after t years. What is the starting balance?
Answer: 1000

6. Write a factored form of a quadratic with roots 3 and -1.
Answer: $(x - 3)(x + 1)$

Grade 11 Practice

Standard F.IF.9

Computation

1. Compare $f(x) = 2x + 1$ and $g(x) = 3x$. Which has the greater slope? Write f or g. **g**

2. Which grows faster in the long run, a linear or an exponential function? Write your an

3. Which has the greater y-intercept, $f(x) = x^2 + 5$ or $g(x) = 2$? Write f or g. **f**

4. Which function is bounded above, $f(x) = -x^2 + 4$ or $g(x) = 2x$? Write f or g. **f**

Word problems

5. Plan A costs \$30 + \$5 per hour and Plan B costs \$10 per hour. Which plan costs less at 8 hours? Write A or B.
Answer: A

6. A ball has height $h = -16t^2 + 64t$. What is its maximum height?
Answer: 64

Grade 11 Practice

Standard F.BF.3

Computation

1. The graph of $y = f(x) + 3$ is shifted how far, and in which direction? Write your answer.

2. The graph of $y = f(x - 2)$ is shifted in which direction? Write right or left. **right**

3. The graph of $y = -f(x)$ is reflected over which axis? Write x or y. **x-axis**

4. The graph of $y = f(2x)$ is compressed in which direction? Write horizontally or vertical

Word problems

5. The graph of $y = x^2$ is shifted left 4 units. Write the new equation.
Answer: $y = (x + 4)^2$

6. The graph of $y = f(x)$ is stretched vertically by 3. Write the new equation.
Answer: $y = 3f(x)$

Grade 11 Practice

Standard F.BF.4

Computation

1. Find the inverse of $f(x) = 2x + 6$. $f^{-1}(x) =$ ___ **$(x - 6)/2$**

2. Find the inverse of $f(x) = x^3$. $f^{-1}(x) =$ ___ **x^(1/3)**

3. Find the inverse of $f(x) = x + 5$. $f^{-1}(x) =$ ___ **$x - 5$**

4. If $f(x) = 2x$, what is $f^{-1}(f(4))$? ___ **4**

Word problems

5. The formula $F = (9/5)C + 32$ converts Celsius to Fahrenheit. Write the inverse formula for C.
Answer: $C = (5/9)(F - 32)$

6. If $f(x) = 3x - 1$, what is $f^{-1}(8)$?
Answer: 3

Grade 11 Practice

Standard F.LE.4

Computation

1. Solve log 8 = ___ **3**

2. Solve log 1000 = ___ **3**

3. Solve log 9 = ___ **2**

4. Solve log 1 = ___ **0**

Word problems

5. Solve 2 = 32 for x.
Answer: 5

6. Solve 10 = 100 for x.
Answer: 2

Grade 11 Practice

Standard F.TF.1

Computation

1. π radians equals how many degrees? ___ **180**

2. π/2 radians equals how many degrees? ___ **90**

3. 2π radians equals how many degrees? ___ **360**

4. Write 60° in radians. ___ **π/3**

Word problems

5. A wheel turns through 90°. Write the angle in radians.
Answer: π/2

6. Write π/4 radians in degrees.
Answer: 45°

Grade 11 Practice

Standard F.TF.2

Computation

1. Find cos 0. ___ **1**

2. Find sin(π/2). ___ **1**

3. Find cos π. ___ **-1**

4. Find sin 0. ___ **0**

Word problems

5. On the unit circle, what point is at angle π/2?
Answer: (0, 1)

6. Find sin(π/6).
Answer: 1/2

Grade 11 Practice

Standard F.TF.5

Computation

1. What is the period of sin x? Write in terms of π. **2π**

2. What is the amplitude of y = 3 sin x? ___ **3**

3. What is the period of y = sin(2x)? Write in terms of π. **π**

4. What is the midline of y = 2 cos x + 5? Write the equation. **y = 5**

Word problems

5. Daylight hours follow h = 12 + 3 sin(t). What is the amplitude?
Answer: 3

6. A tide oscillates between 1 foot and 9 feet. What is its amplitude?
Answer: 4

Grade 11 Practice

Standard S.ID.6

Computation

1. A scatter plot has a positive trend. Write positive or negative for the correlation. **posi**

2. A line of best fit is y = 2x + 1. What is the predicted y when x = 4? ___ **9**

3. Does an outlier affect a regression line? Write yes or no. **yes**

4. A residual is observed minus predicted. Write the word for the difference. **residual**

Word problems

5. A model is y = 0.5x + 10. What is the prediction when x = 20?
Answer: 20

6. A data point is 15 and the model predicts 12. What is the residual?
Answer: 3

Grade 11 Practice

Standard S.ID.7

Computation

1. A line has slope 3. What does this mean for y as x increases by 1? **y increases by 3**

2. In cost = 50 + 2x, what is the fixed cost? ___ **50**

3. A line has slope -1.5. How does y change when x increases by 1? **decreases by 1.5**

4. In y = 40 + 5x, what does the intercept 40 represent? Write starting value or rate. **sta**

Word problems

5. Cost is 20 + 4x, where x is items. What does the slope represent?
Answer: $4 per item

6. Temperature is T = 60 - 2h, where h is hours. What does the slope represent?
Answer: drops 2 degrees per hour